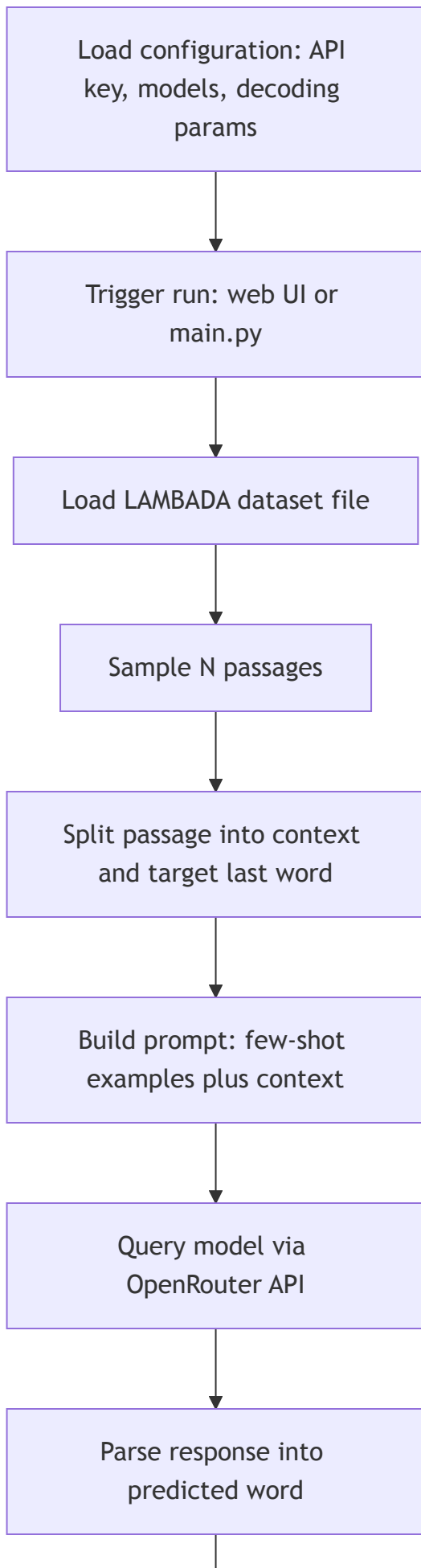
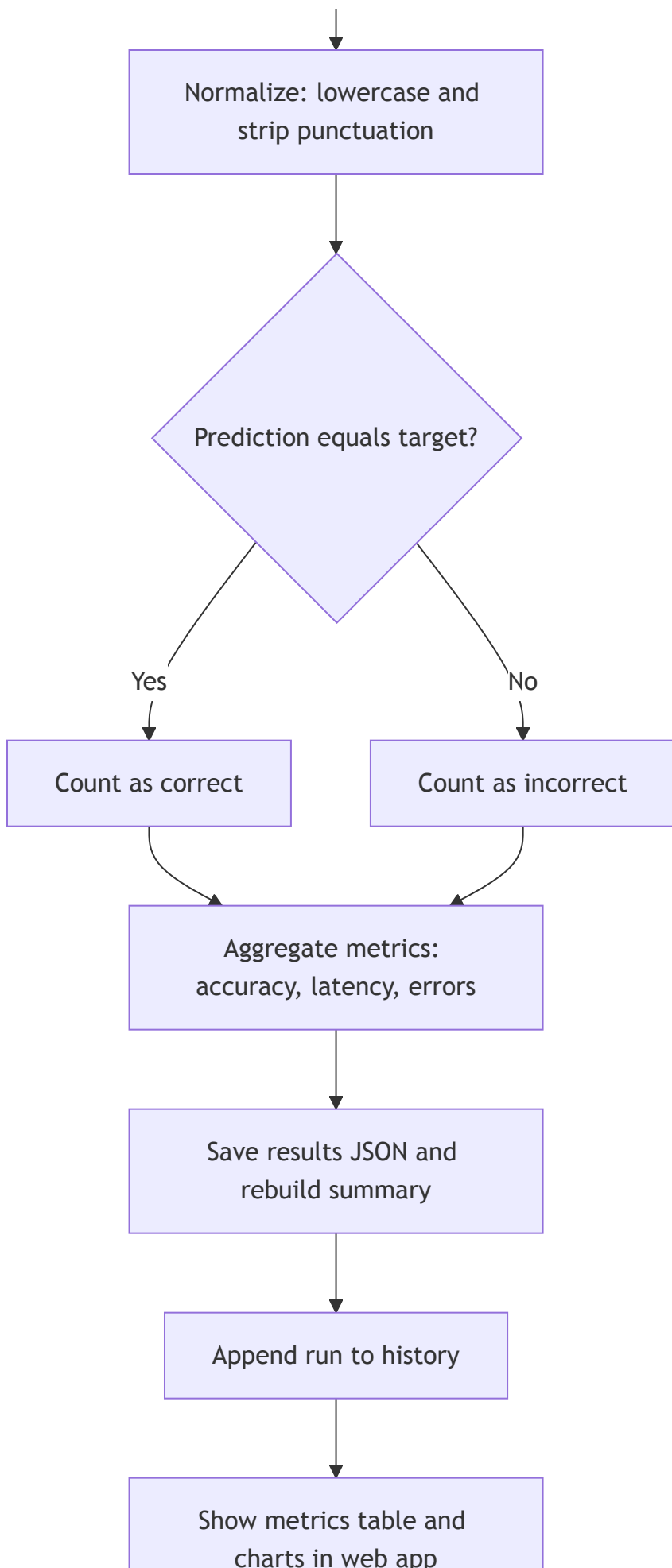


LAMBADA Process Flow

How the benchmark runs in this project, from a click on "Run Benchmark" to the metrics and charts shown in the web app. LAMBADA is a last-word prediction task: the model reads a passage with the final word removed and must predict that word, which is then compared to the target by exact match.

Workflow diagram





Chronological steps

#	Step	What happens	Where (file / function)
1	Configuration	Read API key, model list, decoding params, dataset paths	config.py
2	Trigger	User clicks Run Benchmark (single or all) or runs the pipeline	passenger_wsgi.py (/run , /run_all) or main.py
3	Load dataset	Read the LAMBADA plain-text split; one passage per line	evaluate_lambada.load_dataset
4	Sample	Randomly select N passages (seeded for repeatability)	evaluate_lambada.load_dataset
5	Split	Separate each passage into context and the target last word	load_dataset (rsplit on last space)
6	Build prompt	Prepend few-shot worked examples, then the context	build_few_shot , query_model
7	Query model	Send the prompt to the chosen model through OpenRouter	query_model (HTTP POST)
8	Parse	Extract the single predicted word from the raw response	extract_prediction
9	Normalize	Lowercase and strip surrounding punctuation, both sides	normalize_word
10	Compare	Exact match: predicted word equals target word	evaluate_model

#	Step	What happens	Where (file / function)
11	Aggregate	Tally correct, accuracy, average latency, and errors	<code>evaluate_model</code>
12	Save	Write per-model results JSON and rebuild the summary	<code>run_single_benchmark</code> , <code>rebuild_summary</code>
13	History	Append the run (models, samples, params) to history	<code>append_history</code>
14	Present	Render the metrics table and accuracy/latency charts	<code>index.html</code> (Chart.js); Docs tab renders <code>report.md</code>

Metrics produced

Metric	Definition
Accuracy	correct / total after normalization (exact match)
Average response time	Mean wall-clock seconds per API call
Errors	Count of failed or timed-out API calls
Throughput	total / total wall-clock time (samples per second)

ELI5: Take a story with the last word hidden, ask the model to guess that word, clean up the guess, and check if it matches. Do this for many stories, count how often it is right and how fast it answers, then show the score as a table and bars.